

ENPH 334 Lab Introduction

Teaching Assistants: Hector Hawley Herrera, Hugh Morison,
Nayem Al Kayed, Fraser McCaully, Zhimu Guo (quizzes/exams)

Electronics Technician: Mark Evans

Rules

- No food or liquids inside the lab at any time.
 - Leave water or any liquid containing flasks outside, in your bags or the sink.
 - If you have the need to drink water or eat food, step outside.
- Return all your borrowed parts at the end of the lab session to their corresponding bins.
- Blown fuses, or broken (burnt) parts go into the trash bin (or if you have enough by the end of the semester, make a necklace).
- If you break any benchtop equipment, ~~it will automatically be charged to your SOLUS account~~ let a TA know what is the problem and we will get you a replacement ASAP :P

Objectives of the lab

- Learn the practical and technical skills behind designing and constructing a circuit.
 - Simulation, prototyping, testing, debugging
- Proper handling and use of a multimeter, signal generator and oscilloscope — the bread and butter of an electronics engineer
- Present data in a way that maximizes the utility of the plot or numbers.
- Introduction to research skills to help in the development and design of a circuit from scratch.
- Building and sustaining a good habit of writing a lab notebook.
- Blow as many fuses as possible!

How the labs will work week by week

- You must complete all the **Tasks** in your lab report.
- Attendance for a particular lab is only written down if all the tasks were reviewed by a TA. Your responsibility to make sure you get checked off.
- Raise your hands once a task is completed. Ask questions: task review is a time for TAs to give feedback and correct any misconceptions.
 - If many people have their hands up, move on and raise your hand again when it's calmer.
- If a lab was not completed on time: finish it next week, or in off-hours (**caution**: be aware this means no TA supervision). If you would like access to the lab off-hours either e-mail any of the TAs or Mark Evans)
- Later in the semester there will be extra hours with TA help where you can work on missing weeks.
- There is no deliverables at the end of the week. The goal is the lab completion mark.

ENPH 334: Lab A0

Introduction to Simulation Tools (LTspice)

LTSPICE

- Three Relevant types of simulations:
 - DC Operating Point Simulation
 - Transient (Time-Domain) Simulation
 - AC Sweep (Frequency-Domain) Simulation
- We are using LTSpice in this course
 - First year switching over from OrCAD
 - LTSpice is free and multi-platform
 - Simulation only, no PCB design capability
 - Simpler

